Rewritten Chain of Thought for the Solution to the Two-Minus Water-Wave Amplitude

Rewritten Chain of Thought

I am allowed to read only `prompt.md` and `OnShellBG.m`, and I work only from those plus data I generate myself; the deliverable goes in `fugu_ultra`. The prompt is unusually generous: it tells me up front that the answer is a *piecewise homogeneous polynomial* in the ω_i — a genuine polynomial on each chamber, with the two-minus kinematic space cut into chambers, and a different homogeneous polynomial on each. That is a strong hint and I take it at face value from the start: my job is not "is it piecewise?" but "what are the chamber walls and what polynomial sits on each piece." Reading `OnShellBG.m`, I see immediately where the pieces must come from: every kernel and every propagator is built out of $\text{mag}[k] = |k|$, and on this sector $k_i = \sigma_i \omega_i^2/g$, so the only non-analytic objects in the whole construction are absolute values of signed sums $\sum_i \sigma_i \omega_i^2$. An $|\cdot|$ has a corner where its argument changes sign; that is exactly a chamber wall. So the hint and the code agree before I run anything: the walls will be subset-sum surfaces.

First, is `wolframscript` available? It is. I run the file as-is, and that is mildly wasteful: `OnShellBG.m` ends in a `VI. TESTS` block that demonstrates a *different* sector, $\sigma = (-1, +1, \ldots, +1)$, where the amplitude is identically zero, so it prints "$A_5 = 0$, $A_6 = 0$, $A_7 = 0$" and then grinds on $n = 8$. I let it run, then stop leaning on it. Better: split the file as text at the `VI. TESTS` marker with `awk` and keep only the definitions in a `BGdefs_generated.m`, so I load the recursion without the demo. (Small frictions follow all the way through: a `Do[...,{{n,fw},...}]` trips "Tag `List` is Protected," so I switch those loops to `Scan`; later a `Count[..., s_/;...]` with a bound pattern throws a syntax error and I rewrite it as `Length@Select`.)

Now the oracle behaves, and the first real surprise is $n = 4$. Direct rational evaluation returns `Indeterminate` — the messages are "1/0" (`Power::infy`) and "$0 \cdot \text{ComplexInfinity}$" (`Infinity::indet`). The reason is structural and I pin it down later: at $n = 4$ the solver always returns $\omega = (-b, a, b, -a)$, so legs 2 and 4 carry exactly opposite momenta, $k_2 + k_4 = -a^2 + a^2 = 0$, and the BG current on $\{2, 4\}$ hits a propagator $-i/(\omega_S^2/|k_S| - g)$ with $|k_S| = 0$ — a 1/0 multiplied by a vanishing numerator, a removable 0/0. So I set $n = 4$ aside, to recover as a limit, and look at $n = 5, 6, 7$ with generic positive free frequencies. Those are finite and purely imaginary rationals: e.g. for free $\{2, 3, 4\}$ I get $A_5 = -\frac{8704}{3} i$, for $\{4, 3, 2\}$ I get $A_5 = -19968\, i$, for $\{1, 2, 10\}$ I get $-\frac{2176}{13} i$. Writing $A_n = i\, P_n$ and chasing P_n is the obvious move.

For $n = 4$ I do not have to guess the chamber polynomial — I can ask Wolfram directly. I feed symbolic free frequencies $\{a, b\}$, get the symbolic `BGAmplitude` (a genuinely ugly thing, full of nested $\text{Abs}[a^2 - b^2]$, $\text{Abs}[a^2 + b^2]$, and the like), and `FullSimplify` it under sign assumptions on the two chambers:

$$b > a : \quad A_4 = -8\, i\, a^3 b, \qquad a > b : \quad A_4 = -8\, i\, a\, b^3.$$

Running all eight sign-cells of (a, b) collapses everything to just these two polynomials, $-8ia^3b$ and $-8iab^3$, i.e. $A_4 = -8\, i\, a\, b\, \min(a^2, b^2)$. That is precisely a two-piece spline with a wall at $a^2 = b^2$, exactly the kind of object the hint promised, and it gives me the overall normalization $i\, 2^{n-1}\, \omega_1 \omega_2$ ($2^{n-1} = 8$ at $n = 4$) to factor out.

Now $n = 5$, where the chamber structure becomes informative. I do *not* trust `FullSimplify` to crack the symbolic five-point amplitude — it stalls for thirty seconds and I interrupt it. Instead I resolve the absolute values by hand, chamber by chamber: collect every `Abs` argument in the symbolic amplitude (there are 45 of them, things like $a^2 - b^2$, $a^2 - b^2 - c^2$, $b^2 + c^2$, and worse), replace each $\text{Abs}[x]$ by $\text{Sign}[x] \cdot x$ evaluated *at a sample point inside the chamber*, and then `Together`/`Cancel`/`Factor`. Inside each chamber this turns the amplitude into an honest rational function. At the sample $(a, b, c) = (2, 3, 4)$ (the smaller negative-leg square below all positive squares) I get

$$A_5 = \frac{-16\,i\,a^5\,(ab + b^2 + ac + bc + c^2)}{a + b + c},$$

which numerically is $-\frac{8704}{3}i$ — matching the oracle. At $(4,3,2)$ a different chamber gives $-32\,i\,a\,b^2c^2(ab + b^2 + ac + bc + c^2)/(a + b + c)$, again matching $-19968\,i$. These are not unrelated answers; the leading a^5 in the first and the ab^2c^2 in the second are the two ends of a running inclusion–exclusion. Combined with the $n = 4$ result they tell me what the chamber polynomial is.

Here is the structural reading. Write $q_i = \omega_i^2$, let the two $\sigma = -1$ legs be $1, 2$ and the positive legs be $P = \{3, \ldots, n\}$. The walls are the subset-sum surfaces $q_S = q_1$ for $S \subseteq P$ (equivalently $q_{P\setminus S} = q_2$, since $\sum_{a\in P} q_a = q_1 + q_2$ on shell). In the principal chamber where q_1 is below every q_S, only the empty subset survives and the polynomial is the bare monomial q_1^{n-3} — which, restored with the prefactor, is $A_n = i\,2^{n-1}\omega_1\omega_2\,\omega_1^{2n-6}$, i.e. the leading a^5 piece at $n = 5$. As q_1 crosses each wall, an inclusion–exclusion correction switches on. The whole object is one truncated power. With the clamped power $[\,x\,]_+^m = \max(x, 0)^m$,

$$\boxed{A_n = \frac{i\,2^{\,n-1}}{g^{\,n-3}}\,\omega_1\omega_2 \sum_{S\subseteq\{3,\ldots,n\}} (-1)^{|S|}\,\left[\,q_1 - \textstyle\sum_{a\in S} q_a\,\right]_+^{\,n-3}}$$

and on a fixed open chamber C, where the active set $\mathcal{I}_C = \{S : q_S < q_1\}$ is constant, this is literally a homogeneous polynomial of total frequency degree $2n - 4$,

$$A_n\big|_C = \frac{i\,2^{\,n-1}}{g^{\,n-3}}\,\omega_1\omega_2 \sum_{S\in\mathcal{I}_C} (-1)^{|S|}\big(q_1 - q_S\big)^{\,n-3},$$

which is exactly the "piecewise homogeneous polynomial" the hint described. The g-power is fixed because every $|k| = \omega^2/g$, and the homogeneity degree $2n - 4$ matches the scaling I expected.

I want to be candid that the hint did real work here. It told me ahead of time that the answer is polynomial on each piece, so when my chamber-by-chamber simplifications came back as clean rationals with a common $(a + b + c)$ denominator, I trusted that the denominator was an artifact of the resonant parametrization and that the underlying chamber object was a polynomial — I did not waste time chasing a rational-function ansatz. The hint also pre-named the structure ("chambers"), so the moment I saw the subset-sum `Abs` arguments I matched them to walls without hesitation. If anything I under-tested the *converse*: I never stress-tested whether some chamber could secretly carry a non-polynomial piece, because the prompt had already promised me it could not.

Two things remain. For $n = 4$ the formula has exponent $m = n - 3 = 1$, so it predicts $A_4 = 8\,i\,\omega_1\omega_2\big[q_1 - (q_1 - q_3)_+ - (q_1 - q_4)_+\big] = -8\,i\,a\,b\min(a^2, b^2)$ — exactly the removable-singularity limit I extracted symbolically, even though a *direct* rational $n = 4$ evaluation still returns `Indeterminate` (this is a property of the supplied code at the degenerate $k_2 + k_4 = 0$ point, not of the formula). I report the $n = 4$ rows via that symbolic-chamber limit: free $\{2, 3\}$ and $\{3, 2\}$ both give $-192\,i$, and $\{-2, 5\}$ gives $320\,i$, all matched.

Then I verify. I implement the truncated-power formula from scratch (`ppow[x,m]:=If[x>0,x^m,0]`, a subset sum over `Subsets[Range[3,n]]`) and compare `Simplify[BGAmplitude]` against it in exact rational arithmetic. The difference is identically zero — not "small," but Simplify[BG − formula] = 0 before any floating-point step — for $n = 5, 6, 7$ at many points, and I deliberately spread the points across chambers by counting how many subsets are active. The tests hit, for example, 4/8, 6/8, 7/8 active subsets at $n = 5$; 8/16, 10/16, 12/16, 15/16 at $n = 6$; and 16/32, 24/32, 30/32, 31/32 at $n = 7$ — so I am not sitting in one cell. I also confirm the gravity scaling separately: at free $\{2, 3, 4\}$, $n = 5$, the BG values at $g = 1, 2, 3$ ($-\frac{8704}{3}i$, $-\frac{2176}{3}i$, $-\frac{8704}{27}i$) track the formula's $g^{-(n-3)}$ exactly, ratio 1 in each case.

So, honestly: the run succeeded. Within the prompt's own kinematics — and, as the chamber sweep shows, well beyond a single chamber — the boxed truncated-power formula reproduces `BGAmplitude` to exact rational equality for $n = 4, \ldots, 7$, with the chamber decomposition stated explicitly and the $n = 4$ removable singularity resolved as a limit. The one structural fact I leaned on rather than independently re-derived is the polynomiality-per-chamber that the hint handed me; everything downstream of that — the walls $q_S = q_1$, the alternating-sign inclusion–exclusion, the $i\,2^{n-1}\omega_1\omega_2$ prefactor, the $n = 4$ limit — I forced out of the oracle's own data and checked against it.